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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 11

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Yashmit Kotekar**
PRN No.	**23070122236**
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Docker Compose for Multi-Container App
Theory:	Docker Compose is a tool used to define and manage multi-container Docker applications using a single docker-compose.yml file. It allows multiple services, such as a web application and a database, to be configured, started, and managed together with a single command. Docker Compose automatically creates a network for communication between services, manages named volumes for persistent storage, and simplifies deployment by eliminating the need to start each container individually. This makes it easier to develop, test, and maintain applications that consist of multiple interconnected containers. **Commands Used** • Created a new project directory and navigated into it: mkdir my-multi-app cd my-multi-app • Started the multi-container application in detached mode: docker compose up -d • Verified the running containers: docker compose ps *(or docker ps if docker compose ps was not available)* • Stopped and removed the containers and network created by Docker Compose: docker compose down

	Step 1: Create the Project Directory Created Project Directory — Created a new project directory named my-multi-app and navigated into it to store the Docker Compose configuration. **Step 2: Create docker-compose.yml** Created Docker Compose Configuration — Created and saved the docker-compose.yml file defining the Flask web service, Redis service, Docker volumes, and networking configuration for the multi-container application. **Step 3: Start the Multi-Container Application** Started Multi-Container Application — Started the multi-container application using Docker Compose, which created the required network, volumes, and launched both the Flask web service and Redis service. **Step 4: Access the Web Application** Verified Web Application — Accessed the Flask application through http://localhost:8083 and verified that the web service was successfully running as part of the Docker Compose application. **Step 5: Verify the Running Services** Verified Running Services — Verified that the Docker Compose application was running successfully by listing the active containers, including the Flask web service and the Redis service. **Step 6: Stop and Remove the Multi-Container Application** Stopped Multi-Container Application — Stopped and removed the Docker Compose application, including the running containers and the automatically created network, using the docker compose down command.

docker-compose.yml U

docker-compose.yml

```yaml
version: "3.9"

services:

  db:
    image: postgres:16

    container_name: postgres-db

    restart: always

    environment:
      POSTGRES_DB: mydatabase
      POSTGRES_USER: postgres
      POSTGRES_PASSWORD: postgres

    ports:
      - "5432:5432"

    volumes:
      - postgres_data:/var/lib/postgresql/data

  web:

    build: .

    container_name: flask-app

    depends_on:
      - db

    ports:
      - "5000:5000"

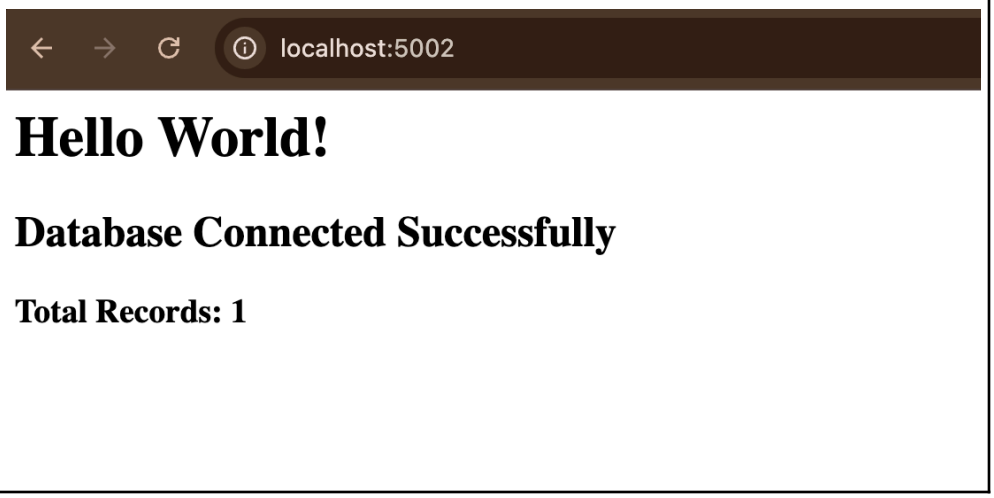
volumes:

  postgres_data:
```

```
yashmitkotekar@Yashmits-MacBook-Air Devops-Coursework % docker compose build
WARN[0000] /Users/yashmitkotekar/Coding/devop/Devops-Coursework/docker-compose.yml: the attribute `version` is obsolete, it will be
 ignored, please remove it to avoid potential confusion
[+] Building 7.8s (12/12) FINISHED
 => [internal] load local bake definitions                                                                                0.0s
 => => reading from stdin 564B                                                                                            0.0s
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 199B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                       4.4s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93  0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93  0.0s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 6.00kB                                                                                       0.0s
 => CACHED [2/5] WORKDIR /app                                                                                             0.0s
 => [3/5] COPY requirements.txt .                                                                                         0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                              2.6s
 => [5/5] COPY . .                                                                                                        0.0s
 => exporting to image                                                                                                    0.6s
 => => exporting layers                                                                                                   0.5s
 => => exporting manifest sha256:15ee10657e736e0128731c068a98f10af06836609925dc4223dea0f645b07b85                         0.0s
 => => exporting config sha256:f1afbddeb49d0ea59fd06bacb376008c10a6a850a650dec22f0b08de5e3d30d9                           0.0s
 => => exporting attestation manifest sha256:2235033fcf042dc0eefadde0fb6af2d494dd1c29977e987c974efc0d0b9f8518             0.0s
 => => exporting manifest list sha256:6cfb6c2b27b538a941b4b0849507ec97af54692f65e0c041cf75962ab3d729c3                    0.0s
 => => naming to docker.io/library/devops-coursework-web:latest                                                           0.0s
 => => unpacking to docker.io/library/devops-coursework-web:latest                                                        0.1s
 => resolving provenance for metadata file                                                                                0.0s
[+] build 1/1
 ✔ Image devops-coursework-web Built                                                                                      7.9s
yashmitkotekar@Yashmits-MacBook-Air Devops-Coursework %
```

```
yashmitkotekar@Yashmits-MacBook-Air Devops-Coursework % docker compose up
WARN[0000] /Users/yashmitkotekar/Coding/devop/Devops-Coursework/docker-compose.yml: the attribute `version` is obsolete, it will be
 ignored, please remove it to avoid potential confusion
[+] up 20/20
 ✔ Image postgres:16                          Pulled                                                                     26.3s
 ✔ Network devops-coursework_default          Created                                                                     0.0s
 ✔ Volume devops-coursework_postgres_data     Created                                                                     0.0s
 ✔ Container postgres-db                      Created                                                                     0.1s
 ✔ Container flask-app                        Created                                                                     0.0s
Attaching to flask-app, postgres-db
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:5000 -> 127.0.0.1:0: listen tcp 0.0.0.0:5000: bind:
address already in use

What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"
yashmitkotekar@Yashmits-MacBook-Air Devops-Coursework %
```

Subject Faculty Name: Prof. Arijit Dutta